



OPEN Trend and change-point dynamics of urban water supply in Gondar City, Ethiopia: a sequential mixed-methods analysis

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Urban water systems in sub-Saharan Africa face increasing pressure due to climate variability, aging infrastructure, and institutional constraints. This study examined the long-term trends and climate responsiveness of water sources supplying Gondar, Ethiopia, using monthly data from the Angereb Reservoir, groundwater wells, and springs (2011–2024). We applied Mann–Kendall tests, change-point detection, and the Standardized Precipitation Index (SPI) with lagged correlations, complemented by interviews and focus groups. The results showed that the reservoir output remained stable ($\tau = 0.048$, $p = 0.369$), whereas groundwater ($\tau = 0.391$, $p < 0.001$) and spring supply ($\tau = 0.256$, $p < 0.001$) increased significantly. Change-point analysis revealed key shifts: surface water decline in July 2013, groundwater expansion in July 2017, and spring integration in June 2019. The SPI lag analysis indicated distinct response times: reservoir supply responded inversely after 11–12 months, groundwater after 3–12 months, and springs after 3 months. Despite increased production, household access did not improve due to population growth and distribution losses. Groundwater offsets the static reservoir output, but unmanaged use poses long-term risks. Incorporating climate-response lags and detected change points into seasonal allocation strategies is vital for strengthening urban water resilience during rapid population growth.

Keywords Change-point analysis, Climate resilience, Ethiopia, Groundwater, Socio-hydrology, Urban water supply

Urban water systems in developing countries are under growing strain from population growth, old infrastructure, climate change, and weak institutions^{1,2}. These pressures reduce service reliability and hit low-income communities hardest, causing irregular supply, more waterborne diseases, and lost work time^{3,4}. Still, the main challenge for urban water security is often not a lack of water itself, but rather weak institutions and aging infrastructure^{5,6}.

This study investigates how these pressures have shifted over time by tracking changes in the types of water sources people use. To clarify, we focus on the climate-sensitive highlands of Gondar, Ethiopia. Through a time-series approach and institutional analysis, we examine when and how residents substitute between water sources and what drives these shifts.

Ethiopian highland cities exemplify acute socio-hydrological challenges, where pronounced seasonal rainfall governs water availability. The Kiremt season (June–August) delivers over 80% of annual precipitation, while Belg rains (March–May) contribute minimally, and Bega dry periods (December–January) yield almost no recharge^{7–9}. These intra-annual extremes drive cyclical reservoir depletion, delayed groundwater recovery, and spring yield fluctuations characteristic of tropical highland hydrology¹⁰. Effective urban water management, therefore, demands temporally adaptive strategies that align infrastructure capacity with hydrological variability^{11–14}.

Gondar City, with a population exceeding 431,000¹⁵, represents a critical case study of highland urban water evolution. The Angereb Reservoir (commissioned in 2002), once the main source of water, is nearing the end of its design life due to chronic sedimentation and treatment limitations. Severe dry-season scarcity (December–

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May) coincides with declines across reservoirs, wells, and springs¹⁶. Reactive source diversification—expanded wellfields and spring rehabilitation—has partially offset reservoir stagnation but introduced new vulnerabilities, including borehole underperformance and spatially fragmented operations¹⁷. This combination of aging infrastructure, climatic extremes, and institutional silos necessitates systematic diagnosis beyond prevailing cross-sectional assessments.

Existing Ethiopian urban water studies found low supply coverage¹⁸, water quality¹⁹, and non-revenue water (NRW) losses⁵. However, these predominantly cross-sectional studies overlook temporal dynamics critical to understanding regime shifts from surface to subsurface dependence²⁰. Prior research lacks systematic integration of: (1) time-series trend analysis of multi-source contributions; (2) change-point detection linking hydrological shifts to management interventions; (3) seasonal decomposition revealing intra-annual adaptation patterns; and (4) institutional interpretation explaining source substitution mechanisms.

This study addresses these gaps through three specific objectives: (1) to quantify long-term and seasonal trends in water supply from Angereb Reservoir, groundwater wells, protected springs, and total distribution outputs using statistical trend analysis; (2) to detect structural change-points associated with climatic extremes, infrastructure interventions, or management regime shifts, complemented by SPI assessment and lagged correlation analysis to evaluate source-specific climate responses; and (3) to interpret socio-institutional mechanisms driving observed hydrological transitions and their implications for sustainable urban water governance.

This integrated socio-hydrological approach generates three novel contributions. Empirically, it provides the first systematic quantification of Gondar's multi-source water evolution (2011–2024). Methodologically, it establishes a sequential mixed-methods framework combining advanced time-series analytics with institutional analysis. Theoretically, it documents path-dependent adaptation from surface water dominance to subsurface vulnerability, revealing how institutional responses to physical decline reshape urban water architecture under highland constraints.

Materials and methods

Study area

Gondar City lies approximately 750 km north of Addis Ababa in northwestern Ethiopia at about 2200 m above sea level (Fig. 1). The city experiences pronounced rainfall seasonality, with over 80% of annual precipitation occurring from June to August, followed by an extended dry period from October to May²¹. Mean annual rainfall is approximately 1200 mm, with mean monthly temperatures ranging from 16 to 24 °C.

The current water supply system of Gondar city comprises three main source types. The Angereb Reservoir, created in 2002, is the primary surface water source with an original design capacity of 5.3 million cubic meters²². However, sedimentation has reduced live storage by an estimated 40% since commissioning²³. Groundwater is extracted from thirteen operational boreholes: eight in the Angereb Well Field and five high-yield wells in the Kola Diba Well Fields commissioned in 2017. Two low-yield springs serve local needs but are not part of the

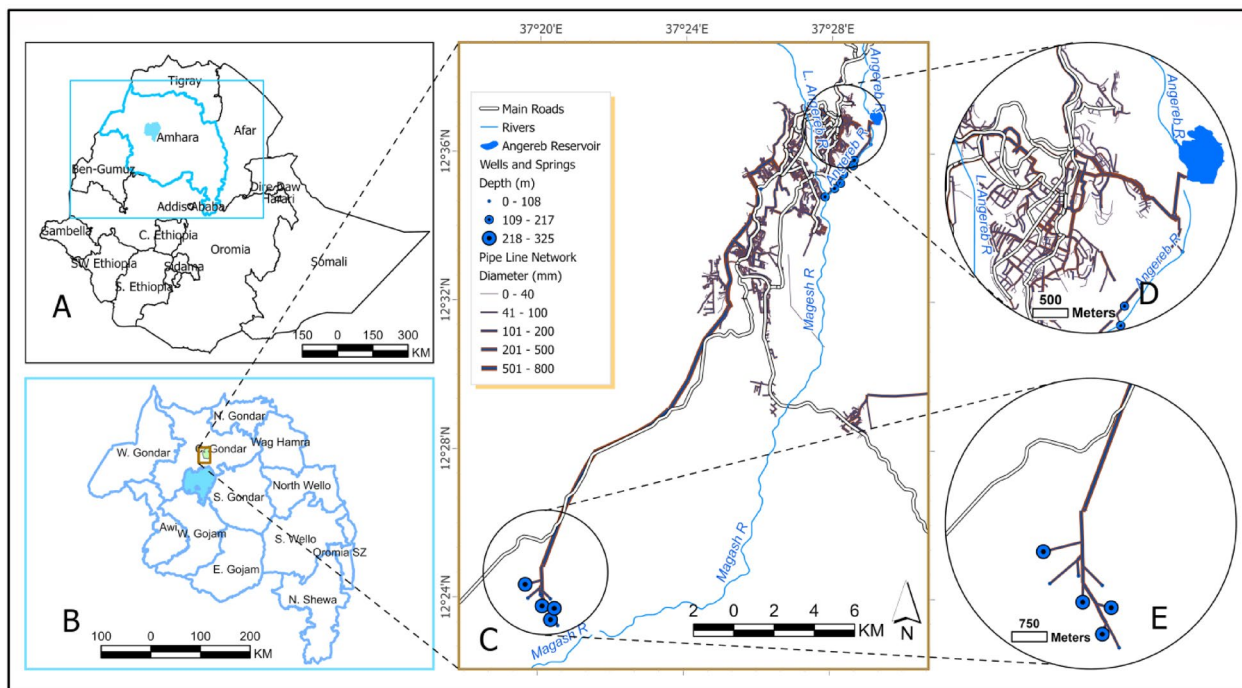


Fig. 1. Location and water supply infrastructure of Gondar City: (a) Ethiopia and the Amhara Region; (b) Amhara Regional States of Ethiopia with Gondar City highlighted; (c) major water sources and pipelines; (d) Angereb Reservoir area; (e) Kola Diba wellfield area.

main distribution system. Together, these sources supply an estimated 1.8 million cubic meters annually, though actual production falls short of growing demand, particularly during the dry season from December to May¹⁷.

Research design

This study employed a sequential explanatory mixed-methods design²⁴, beginning with quantitative analysis followed by qualitative interpretation. This approach first identified long-term patterns and sudden changes through statistical analysis, then interpreted these findings through stakeholder perspectives. Figure 2 summarizes the analytical framework, outlining the phases, inputs, methods, outputs, and points of integration.

Data collection

Monthly water supply records (January 2011–March 2024, 14 years) were obtained from the Gondar City Water Supply and Sanitation Service (GCWSSS). The period begins with the earliest consistently digitized, quality-assured abstraction data; earlier records are fragmented with > 15% missing values. The resulting 168 monthly observations give sufficient length for robust trend detection, given the pronounced seasonal signals. All quantitative datasets are available as Supplementary Table S1 online.

The records covered five primary variables (units: $\text{m}^3 \text{ month}^{-1}$): raw water from the Angereb Reservoir (Rwater), treated water from treatment plant (Twater), groundwater abstraction (Wwater), spring contributions (Swater), and total water supply for distribution (TWS). The total water supply (TWS) is the sum of treated water, groundwater, and spring contributions ($\text{TWS} = \text{Twater} + \text{Wwater} + \text{Swater}$). The corresponding climatic variables, monthly rainfall (mm) and mean temperature ($^{\circ}\text{C}$), were obtained from the National Meteorological Agency, 2024 for the Gondar City, Azezo Station.

Qualitative data were collected from April 15 to 20, 2024, through six key informant interviews (KIIs) and four focus group discussions (FGDs) involving 24 participants. Key informants were purposively selected based on: (1) direct involvement in Gondar's water supply management; (2) current or recent operational or policy responsibility; (3) institutional representation (GCWSSS, treatment plant, regional water bureau); and (4) willingness to participate in the study.

Data processing

All quantitative datasets were assessed for missing values, outliers and inconsistencies. Minor gaps (less than 3% of observations) resulting from recording lapses were addressed using linear interpolation to ensure the continuity of the data. Anomalies attributable to documented operational disruptions, including maintenance shutdowns and power outages, were retained to reflect the actual system behavior. For analyses requiring cross-source comparability, the data were harmonized to monthly intervals, converted to volumetric units (m^3), standardized to continuous time indices, and normalized via z-score transformation.

Data analysis

Quantitative analysis was conducted in R (Posit Cloud) using the tidyverse, zoo, trend, and change-point packages. Long-term trends were analyzed using the Mann–Kendall test and Sen's slope estimator. Seasonal variation was assessed using the Seasonal Mann–Kendall test. Significant shifts were detected using the Pettitt, PELT, and CUSUM methods. Climate responses were evaluated using the Standardized Precipitation Index (SPI) and lagged correlation analysis. The causes of the observed patterns were explored via thematic analysis of key informant interviews (KIIs) and focus group discussions (FGDs).

These methods were selected for their suitability in analyzing time-series water data. The non-parametric Mann–Kendall test is robust to non-normality^{25,26} and can be modified to correct for serial correlations using whitening and autocorrelation method^{27,28}. Sen's slope provides robust estimates of trend magnitude with confidence intervals^{29,30}. The SPI captures multi-timescale drought conditions³¹, lagged correlations reveal response timing³¹, and change-point methods identify regime shifts^{32–34}.

Qualitative data analysis:

Qualitative data were analyzed using inductive thematic coding, following established procedures for mixed-methods research³⁵. Interview transcripts were read iteratively to identify recurring patterns related to water source reliability, institutional performance, and local coping strategies. The coding process proceeded in three stages: (1) open coding, which involved identifying initial concepts emerging from the data; (2) axial coding, in which related concepts were organized into broader analytical categories; and (3) selective coding, through which core themes were identified and refined.

Emergent themes were subsequently triangulated with quantitative findings to contextualize statistical results within broader socio-institutional and water management dynamics. A triangulation strategy was employed to enhance analytical rigor and minimize potential bias by systematically comparing accounts across multiple informants, as well as between interview narratives and relevant documentary sources.

Ethics approval and consent to participate

The Research Ethics Committee of the Department of Geography and Environmental Studies, Faculty of Social Sciences, Bahir Dar University, Ethiopia, waived formal ethical review and written informed consent due to the study's low risk and compliance with ethical regulations (Ref. No. Geo/792/2016, approved September 15, 2024). Nonetheless, rigorous ethical measures were implemented: all participants received verbal explanation of the study purpose, procedures, and their rights; participation was voluntary; confidentiality and anonymity were guaranteed; and data were stored securely with access restricted to the research team.

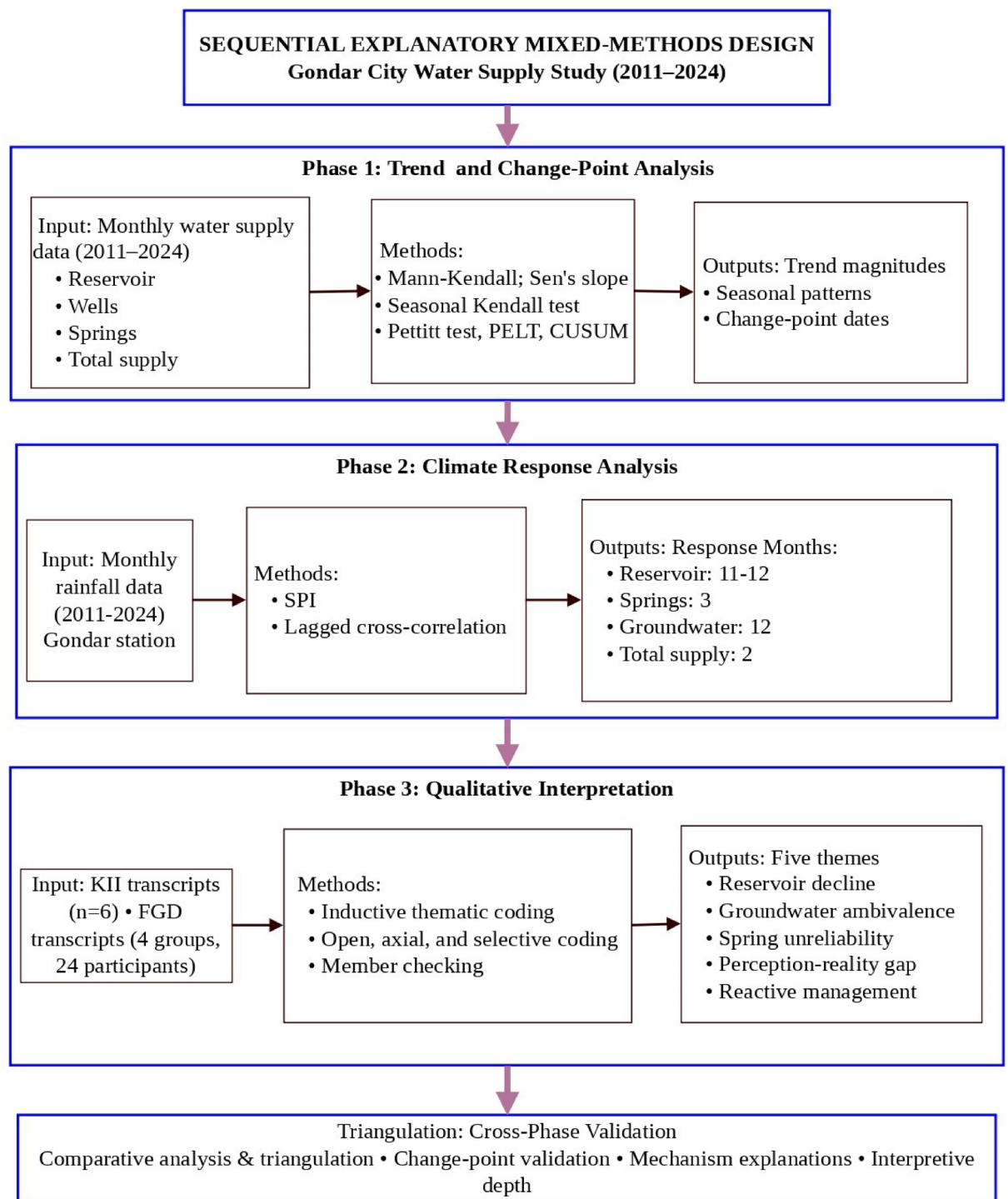


Fig. 2. Sequential explanatory mixed-methods analytical framework. Phase 1: Quantitative trend analysis (Mann–Kendall and Sen's slope). Phase 2: Change point and climate response analysis (Pettitt, PELT, CUSUM, SPI, lagged correlations). Phase 3: Qualitative interpretation (thematic analysis of the KIIs and FGDs). Cross-validation across phases enables integration and stakeholder interpretation of statistical results.

Compliance with ethical guidelines

This study was conducted in accordance with international and national ethical standards for research involving human participants. The Declaration of Helsinki (2013) establishes ethical principles for research involving human subjects, including requirements for informed consent, risk minimization, and privacy protection.

Source	N	Mean	Median	SD	Variance	Min	Max	Range	CV
Rwater	168	269,604	277,336	37,284	1.39×10^9	150,776	367,384	216,608	0.138
Twater	168	253,580	260,696	34,913	1.22×10^9	141,729	345,341	203,612	0.138
Wwater	168	83,141	52,152	53,741	2.89×10^9	20,022	189,806	169,784	0.646
Swater	168	3580	2642	2687	7.22×10^6	377	13,345	12,968	0.751
TWS	168	338,005	318,234	65,012	4.23×10^9	203,141	513,600	310,459	0.192

Table 1. Summary statistics for water supply sources (2011–2024) ($\text{m}^3 \text{ month}^{-1}$). CV, Coefficient of variation (SD/Mean). All values are monthly volumes in cubic meters.

Variable	T	P	Z	Sen's Slope ($\text{m}^3 \text{ month}^{-1}$)	Lower 95% CI	Upper 95% CI
Rwater	0.048	0.369	0.898	563.5	−704.7	1917.40
Twater	0.045	0.398	0.846	492.3	−672.6	1764.00
Wwater	0.391	<0.001	7.387	9078.40	7678.00	10,340.70
Swater	0.256	<0.001	4.825	202.9	115.7	299.6
TWS	0.471	<0.001	8.902	11,414.10	9645.60	13,173.80

Table 2. Mann–Kendall trend test results by source. Kendall's Tau (τ) measures the strength and direction of the monotonic trend in monthly abstraction over time. Z is the standardized Mann–Kendall test statistic. Sen's slope ($\text{m}^3 \text{ month}^{-1}$) estimates the magnitude of the monthly change, with 95% confidence intervals. $P < 0.05$ indicates a statistically significant trend.

Results
Descriptive overview of water supply sources

Five primary indicators characterize water supply of Gondar City for 2011–2024: raw water from Angereb Reservoir (Rwater), treated water (Twater), groundwater extraction (Wwater), spring contributions (Swater), and total water supply to distribution (TWS). Summary statistics (Table 1) reveal marked differences in variability and central tendency across sources.

Water supply from Angereb reservoir (Rwater and Twater) exhibit remarkably stable averages and low variation ($CV \approx 0.14$), reflecting consistent, infrastructure-driven supply. In contrast, supplemental well and spring sources have high coefficients of variation ($CV > 0.6$), indicating substantial operational fluctuation and responsive, adaptive use during shortages. These descriptive patterns suggest a structural imbalance: steady but stagnant surface supply supplemented by highly variable subsurface sources.

Autocorrelation structure and pre-whitening

Time-series diagnostics confirmed significant lag-1 autocorrelation across all sources, necessitating correction before trend analysis. Rwater exhibited AR (1) structure with $\phi_1 = 0.71$ ($\rho = 0.72$). Wwater required AR(2) correction ($\phi_1 = 0.82$, $\phi_2 = -0.15$; $\rho = 0.85$), reflecting short-term memory effects induced by intermittent operation of wells; which are not pumped continuously but activated during shortages, creating a pattern where high abstraction in one month often predicts reduced abstraction in subsequent months as storage recovers. Swater showed AR (1) structure ($\phi_1 = 0.58$; $\rho = 0.58$), and TWS exhibited AR (1) structure ($\phi_1 = 0.68$; $\rho = 0.68$). After pre-whitening, residuals showed no significant autocorrelation (Ljung–Box $p > 0.05$ for all series: Rwater $p = 0.51$, Wwater $p = 0.38$, Swater $p = 0.67$, TWS $p = 0.42$), ensuring unbiased application of the modified Mann–Kendall test.

Long-term trends (2011–2024)

Surface water from the Angereb Reservoir remained consistent throughout the study period, whereas contributions from groundwater and springs increased substantially. Table 2 presents Mann–Kendall trend test results for each source, with p -values indicating statistical significance and Sen's slopes showing the magnitude and direction of change.

Water supply from the Angereb reservoir showed no significant long-term trends (Rwater: $\tau = 0.048$, $p = 0.369$; Twater: $\tau = 0.045$, $p = 0.398$). Sen's slope for Rwater was $563.5 \text{ m}^3 \text{ month}^{-1}$, but the 95% confidence interval crossed zero (−704.7 to 1917.4), indicating that the apparent increase was not significant. This stagnation aligns with sedimentation documented in earlier studies^{23,36}, which estimated approximately 3% annual storage loss, and recent engineering assessments that reported a 40% reduction in live storage since commissioning²³. Key Informants (KIs) confirmed this interpretation; one engineer explained, “The reservoir is like a dying battery. It holds less water every year. The reservoir that served us for 15 years is now only half as useful” (KI-2). Another added, “By 2013, we knew the reservoir would never fully recover. That is when we started looking seriously at wells” (KI-1).

In contrast, groundwater abstraction showed a strong and statistically significant upward trend (Wwater: $\tau = 0.391$, $p < 0.001$; Sen's slope = $9,078 \text{ m}^3 \text{ month}^{-1}$ [95% CI: 7678–10,341]), representing more than double the production. Key informants attributed this increase to the commissioning of the Kola Diba wellfield around

2017 while also expressing sustainability concerns. One official noted: “The wells saved us, but now we worry about them drying too. Some are already showing declining yields.” (KI-5).

Spring contributions also increased significantly (Swater: $\tau=0.256$, $p<0.001$; Sen's slope = $203 \text{ m}^3 \text{ month}^{-1}$ [95% CI: 116–300]), reflecting focused maintenance and rehabilitation that began in 2019. However, a focus group participant observed: “The springs come and go. After good rain, they flow, but during dry periods, they disappear. You cannot rely on them” (FGD-3).

The total water supply increased significantly (TWS: $\tau=0.471$, $p<0.001$; Sen's slope = $11,414 \text{ m}^3 \text{ month}^{-1}$ [95% CI: 9646–13,174]). However, when normalized by population, the per capita supply showed no significant trend. According to CSA projections, the population grew from 307,000 in 2011 to 431,000 in 2024, an increase of approximately 40%, indicating that supply gains have been absorbed by population growth rather than improved per capita access.

Qualitative findings revealed a gap between perceptions and reality. Despite statistical increases in the total supply, residents consistently reported no improvement in household access. One resident summarized this sentiment as follows: “The numbers may show more water, but my tap still runs dry for days. Where is this water going?” (FGD-3). Another noted: “There are more people, but the same amount of water reaches my area. Of course, it is worse” (FGD-2). Key informants estimated non-revenue water at 35–40% (KI-1, KI-4), meaning that substantial production gains were lost to leaks and illegal connections before reaching consumers.

In Gondar, water supply increment is primarily driven by groundwater and springs, whereas reservoirs no longer contribute to system expansion. However, these production gains have not translated into improved household access due to rapid population growth and distribution system losses. Figure 3 presents time series plots for all sources with Sen's slope lines, visually confirming the stagnation in surface water and increases in subsurface sources.

Seasonal variability of water supply

Seasonal patterns reveal critical vulnerability during winter and spring, when groundwater extraction intensifies to compensate for reservoir and spring water declines. Seasonal Mann–Kendall analysis (Table 3) demonstrates complex intra-annual variations consistent with distinct rainfall regimes of Ethiopia (see Supplementary Table S2 online for complete analysis). Sen's slope values were expressed as the average change ($\text{m}^3 \text{ season}^{-1}$) over the study period.

Winter (December to February) The total water supply increased significantly ($p<0.05$), driven by high well pumping (Wwater: $+8,687 \text{ m}^3 \text{ season}^{-1}$; $p<0.05$) and increased spring contributions (Swater: $+347 \text{ m}^3 \text{ season}^{-1}$; $p<0.05$). Despite these increases, KIs reported severe declines of Rwater and Twater, forcing households to rely heavily on alternative sources of water. The focus group participants described winter as the most difficult period to access water. One resident explained, “December to February, we know the taps will be dry. We prepare by storing water in containers, but it is never enough” (FGD-1). This indicates that the statistical increases reflect crisis-driven over-extraction rather than reliable improvements in supply.

Spring (March to May) This period is the most vulnerable period for the system. Total supply rose significantly ($+9218 \text{ m}^3 \text{ season}^{-1}$; $p<0.01$), driven entirely by increased groundwater use ($+9554 \text{ m}^3 \text{ season}^{-1}$; $p<0.05$), whereas spring flows stagnated. KIs confirmed that spring water often diminishes or ceases by late spring, with reservoir evaporation worsening the deficits. Households depend heavily on wells and expensive informal vendors for water supply. A female participant noted, “In spring, we buy water from vendors. It is expensive, sometimes half our daily income, and we never know if it is clean” (FGD-4). This highlights the unsustainable nature of groundwater use during the pre-wet season in the region.

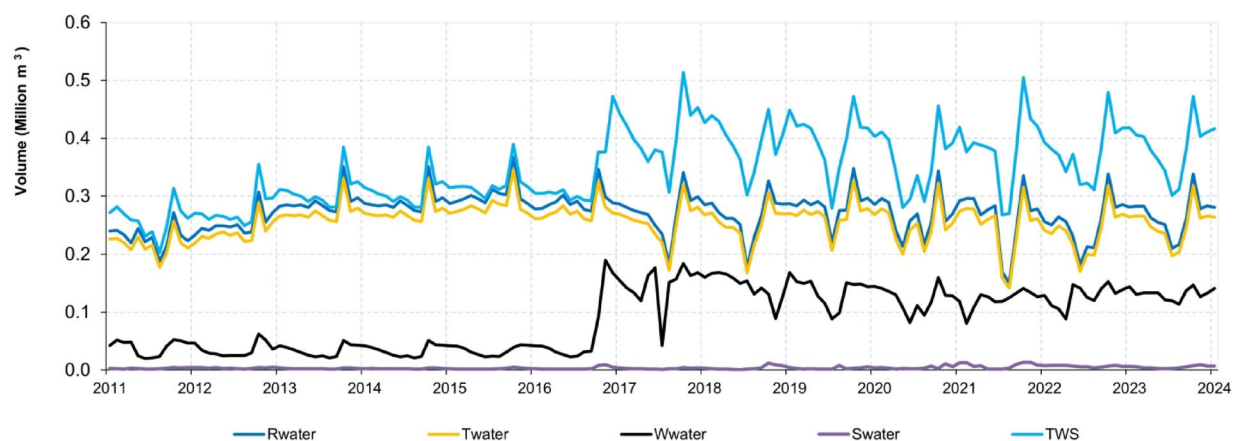


Fig. 3. Long-term monthly water supply trends in Gondar City (2011–2024) Time-series plots for (a) reservoir water (Rwater), (b) treated water (Twater), (c) groundwater (Wwater), (d) spring water (Swater), and (e) total water supply (TWS). Sen's slope lines show stagnation in surface water sources and increasing trends in groundwater, spring water, and total supply.

Season	Variable	T	P	Z	Sen Slope (m ³ season ⁻¹)	Lower 95% CI	Upper 95% CI
Winter	Wwater	0.462	0.025	2.248	8686.6	520.2	12,993.3
	Swater	0.44	0.033	2.138	346.5	68	623.4
	TWS	0.495	0.016	2.409	11,178.3	5479.3	16,430.3
Spring	Wwater	0.487	0.024	2.266	9554.4	191.1	15,162.9
	TWS	0.603	0.005	2.812	9218.1	4513.5	16,859.7
Summer	Wwater	0.484	0.018	2.358	8800.4	1529.8	13,248.7
	Swater	0.681	<0.001	3.344	262.4	127.3	372.5
	TWS	0.571	0.005	2.796	8474.8	3313.2	15,005
Autumn	Swater	0.549	0.007	2.687	435	151.6	735.3
	TWS	0.725	<0.001	3.518	13,163.9	9265	17,459.8

Table 3. Seasonal trends by quarter (significant results only). τ is Kendall's tau indicating the strength and direction of the seasonal trend. $P < 0.05$ indicates significance, Z is the standardized statistic, and Sen's slope (m³ season⁻¹) shows the average seasonal change. 95% confidence intervals excluding zero indicate significant trends. Only significant results are shown.

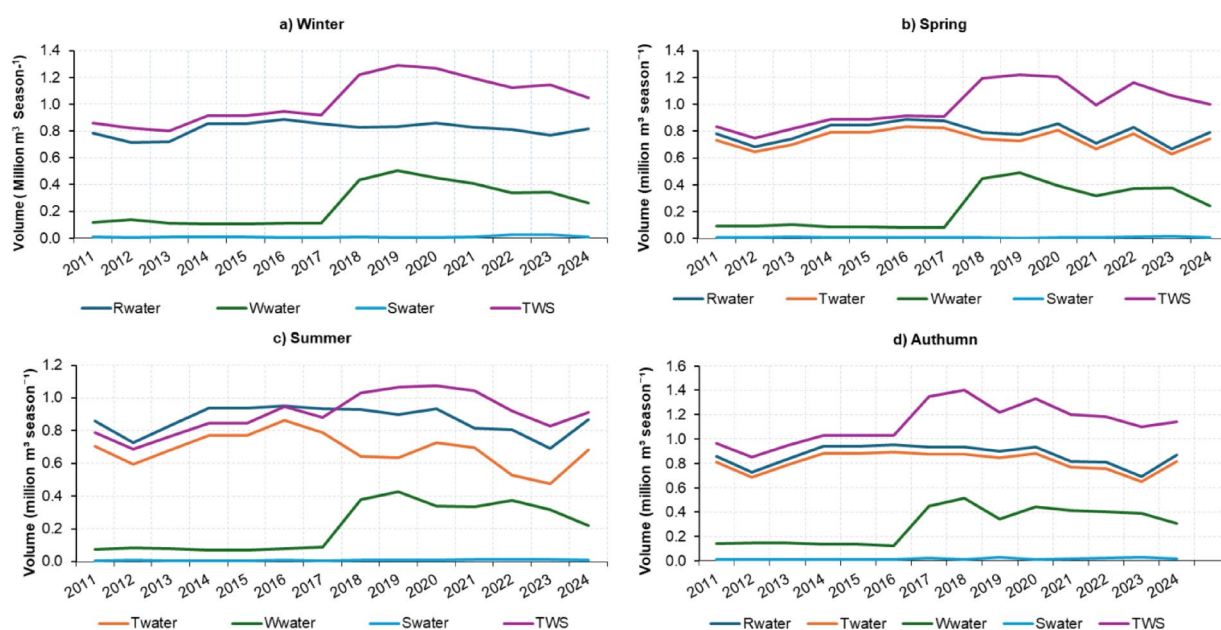


Fig. 4. Seasonal water supply contributions in Gondar City (2011–2024). Quarterly trends from reservoir, groundwater, and spring sources during (a) Winter (Dec–Feb), (b) Spring (Mar–May), (c) Summer (Jun–Aug), and (d) Autumn (Sep–Nov). Significant seasonal trends highlight increased groundwater reliance in dry seasons and spring recovery in wet seasons.

Summer (June to August) peak wet-season water availability was evident from significant increases in spring water (+262 m³ season⁻¹; $p < 0.001$) and total supply (+8475 m³ season⁻¹; $p < 0.01$). Rooftop rainwater harvesting temporarily reduces household demand. However, heavy rains cause sedimentation and operational challenges at Angereb Reservoir intake. A treatment plant manager explained, “When the heavy rains come, the reservoir water turns to mud. We have to reduce treatment or shut down entirely” (KI-3).

Autumn (September to November) this period showed the strongest recovery, with the largest total supply increase (+13,164 m³ season⁻¹; $p < 0.001$), primarily from spring recharge (+435 m³ season⁻¹; $p < 0.01$). KIs described this as a temporary recovery phase, allowing for partial groundwater replenishment. Households and managers strategically use this period to prepare for the upcoming dry season.

Seasonal analysis reveals a recurring crisis-response cycle: Water scarcity during winter season causes excessive groundwater extraction; spring represents the period of greatest vulnerability; summer offers temporary relief but introduces operational difficulties; and autumn allows for brief recovery before the cycle begins again. Figure 4 depicts the seasonal trends for each source across all seasons.

Figure 4 shows seasonal trend plots for each source across the four quarters.

Rainfall variability and water supply responses

Gondar experienced severe drought (2011–2015) and sustained wet conditions (2016–2024). SPI analysis revealed this hydroclimatic shift: from 2011 to 2015, severe to extreme drought persisted, with SPI-12 and SPI-24 values falling below -2 during the peak 2015 El Niño drought period. From 2016 onward, the conditions shifted to near-normal and wet, with a prolonged wet period from 2021 to 2024 (SPI-12 exceeding 1.5 for multiple months; see Supplementary Table S3 and Fig. 5).

Several key relationships emerge between the water sources and rainfall patterns.

Lagged correlation analysis (Table 4) revealed four distinct precipitation-response patterns. First, Angereb reservoir releases were inversely related to rainfall, with higher precipitation predicting reduced withdrawals after an 11-month lag ($R_{\text{water}}: r = -0.454, p < 0.001$). This reflects operational management rather than direct inflow effects, as managers conserve storage when alternative sources are available. One key informant explained: “When the rains are good and springs flow, we rest the reservoir. It is our only buffer for the next dry period” (KI-1).

Second, spring discharge was strongly correlated with recent rainfall at a 3-month lag ($S_{\text{water}}: r = 0.568, p < 0.001$), indicating a rapid response via shallow groundwater flow paths. Third, groundwater abstraction correlated with long-term precipitation at a 12-month lag ($W_{\text{water}} \text{ with SPI-24: } r = 0.432, p < 0.001$), reflecting slower recharge processes in deeper aquifers. Fourth, total water supply correlated positively with rainfall at a 2-month lag ($TWS: r = 0.419, p < 0.001$), driven by climate-sensitive springs and operational groundwater pumping, rather than managed reservoir releases.

These distinct response times—11 months (inverse) for reservoirs, 3 months for springs, 12 months for groundwater, and 2 months for total supply—provide critical information for seasonal operations planning. Managers can anticipate spring contributions approximately one season after rainfall, whereas groundwater responses require multi-year moisture accumulation.

Data limitations Groundwater response was assessed using abstraction data rather than water levels, as no monitoring infrastructure exists in Gondar, a critical gap identified in this study. While the significant correlation between SPI-24 and W_{water} at a 12-month lag is consistent with the recharge response, direct water level measurements are needed for confirmation (see Section 4.1, Recommendation 1).

In seasonal operations planning, understanding response times is critical. Managers can anticipate spring contributions within three months of rainfall. However, groundwater recharge requires 12 months of moisture

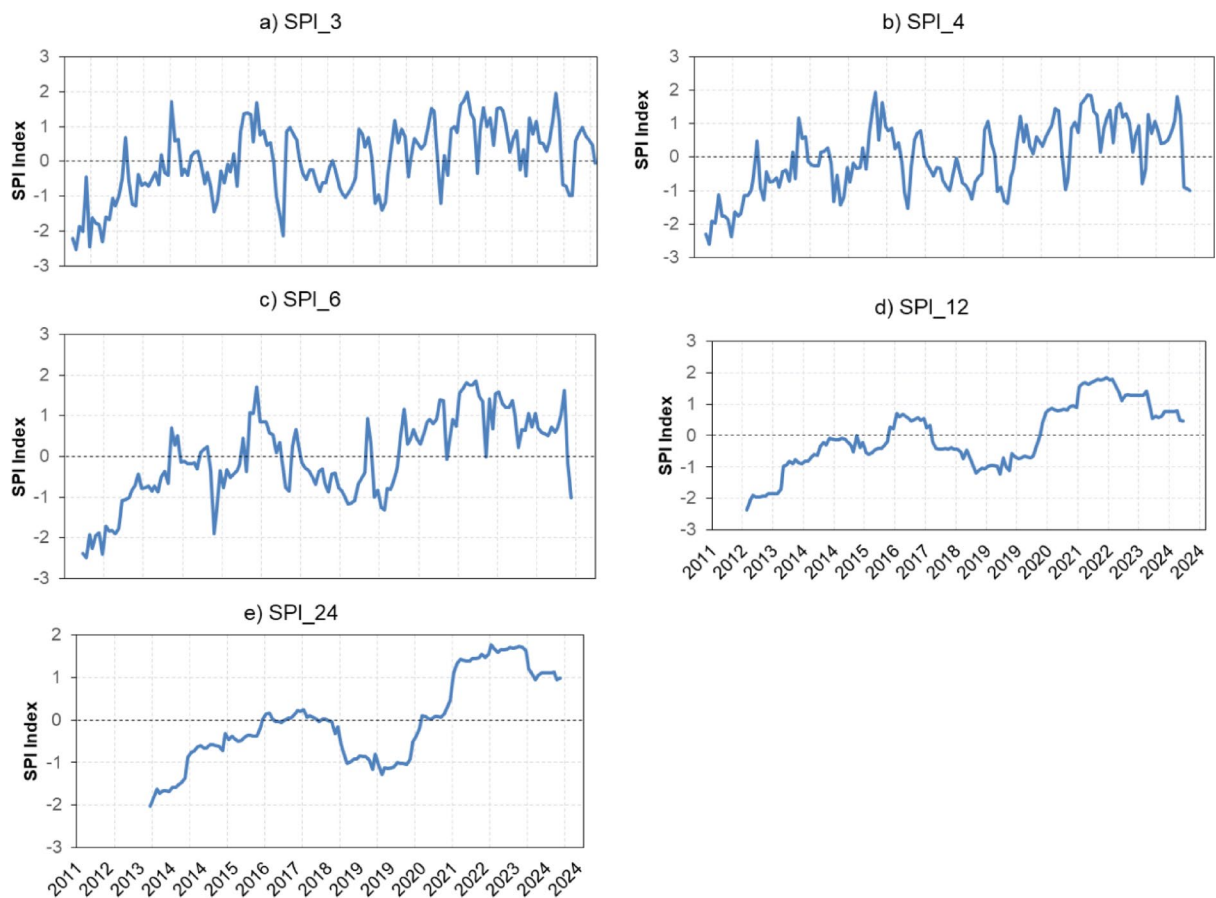


Fig. 5. SPI for Gondar City (2011–2024). SPI values at (a) 1-month, (b) 3-month, (c) 6-month, (d) 12-month, and (e) 24-month scales. The plots show a shift from drought conditions before 2016 to predominantly wet conditions after 2016, with SPI-12 and SPI-24 capturing long-term trends.

Variable	Predictor	Best Lag (months)	Pearson r	p-value	Response type
Rwater	Rainfall	11	-0.454	4.84×10^{-9}	Inverse (managed)
	SPI-24	11	-0.304	4.89×10^{-4}	
	SPI-12	12	-0.207	1.44×10^{-2}	
Twater	Rainfall	11	-0.456	3.98×10^{-9}	Inverse (managed)
	SPI-24	11	-0.304	4.89×10^{-4}	
	SPI-12	12	-0.208	1.41×10^{-2}	
Wwater	SPI-24	12	0.432	4.06×10^{-7}	Slow positive (recharge)
	SPI-12	12	0.409	5.85×10^{-7}	
	Rainfall	3	0.231	3.35×10^{-3}	
Swater	Rainfall	3	0.568	5.49×10^{-15}	Fast positive
	SPI-24	3	0.499	6.55×10^{-10}	
	SPI-12	12	0.449	3.06×10^{-8}	
TWS	Rainfall	2	0.419	3.56×10^{-8}	Mixed (2-month aggregate)
	SPI-24	12	0.278	1.57×10^{-3}	

Table 4. Lagged correlations between climate variables and water supply.

Variable	Pettitt Test	p-value	PELT change-points
Rwater	41,456	1.92×10^{-5}	July 2013, March 2018
Twater	41,456	2.56×10^{-5}	July 2013, March 2018
Wwater	42,917	$< 2.2 \times 10^{-16}$	July 2017, August 2018, April 2019
Swater	43,617	7.16×10^{-5}	Nov 2016, July 2017, Oct 2021, Dec 2021
TWS	42,917	$< 2.2 \times 10^{-16}$	Aug 2012, July 2013, July 2017

Table 5. Change-point detection results. Pettitt test statistics indicate change magnitude, with larger values showing stronger shifts. P-values (e.g., $1.92 \times 10^{-5} = 0.000019$) show statistical significance, with values < 0.05 considered significant. The PELT algorithm identifies dates of significant regime shifts.

accumulation. Conversely, the reservoir's 11-month response time reflects strategic conservation efforts, not hydrological delays.

Change-point and regime shift analysis

Key change points indicated fundamental shifts in water supply of Gondar City. Pettitt and PELT tests, supported by CUSUM plots, identified major turning points in reservoir variables in July 2013, groundwater and total supply in July 2017, and spring water in June 2019 (Table 5; Fig. 6). These change points exceeded the established CUSUM control limits ($CV90 = \pm 14.64$, $CV95 = \pm 16.32$, $CV99 = \pm 19.56$), validating the significant regime shifts identified by the Pettitt and PELT tests.

These change points align with specific documented events: reservoir decline in July 2013, which led to drought-induced intake exposure and pumped abstraction; groundwater expansion in July 2017, coinciding with the commissioning of the Kola Diba wellfield; and spring increase in June 2019, which followed rehabilitation programs. Since 2017, the system has operated under a new regime that is increasingly dependent on groundwater.

Contextual interpretation of change points

The detected change-points correspond to documented operational, climatic, or institutional events, confirming that the statistical breaks reflect actual transformations in the water supply system.

Quantity of water supply from Angereb Reservoir declined in July 2013 after the 2011–2012 droughts, exposing the Reservoir's intake structure and necessitating a shift from gravity-fed to pumped abstraction. Conversely, the July 2017 increase in groundwater and total water supply coincided with the commissioning of the Kola Diba wellfields, comprising five high-yield production wells, and a policy shift prioritizing groundwater development. Similarly, the increase in spring discharge observed in June 2019 followed the completion of a spring rehabilitation program and improved recharge from the wet period that began in 2016.

Additional change points detected by PELT align with documented operational events. Specifically, March 2018 corresponds to a major treatment plant maintenance shutdown and power interruptions, as recorded in GCWSSS maintenance logs, and August 2018 aligns with the installation of new pumps at the Angereb wellfield, documented in procurement records.

The KIs confirmed these interpretations. Regarding the 2013 reservoir change: “By mid-2013, we realized the reservoir would never fully recover and that the intake pipes were exposed. We had to install pumps just to draw water, which changed everything” (KI-2). On the 2017 groundwater expansion, one farmer said: “When Kola



Fig. 6. Cumulative sum (CUSUM) control charts of water supply sources (2011–2024). Charts for (a) reservoir water (Rwater), (b) groundwater (Wwater), (c) spring water (Swater), and (d) total water supply (TWS). Solid lines show cumulative deviations from the mean, with horizontal dashed lines indicating 90%, 95%, and 99% control limits. Crossings of these limits indicate significant change-points, with key events annotated.

Diba came online in 2017, we suddenly had more water than ever before. It was a game-changer, but also a worry, as we did not really know how much we could safely extract” (KI-5).

Following 2017, a new operational regime emerged: surface water stabilized at reduced levels, whereas groundwater and springs expanded, sustaining the overall output. However, one informant cautioned that, “We are surviving, not managing. Every year, we scramble, drilling another well or repairing a spring, but there is no comprehensive plan” (KI-1).

Change-point analysis identifies key regime shifts in the water supply system of Gondar City. A reservoir change in 2013 signaled declining surface water reliability, whereas a 2017 shift toward groundwater reflected a deliberate institutional transition. The increased water supply from springs in 2019 indicates reactive supply diversification. Since 2017, the system has operated under a new regime characterized by an increased dependence on groundwater.

These events were identified through key informant interviews (KI-1 to KI-5) and verified using GCWSSS operational records, project reports, maintenance logs, procurement documentation and published literature²⁶.

Integrated qualitative findings: thematic analysis

To better understand the quantitative trends and provide context on water supply challenges, interview and focus group transcripts were analyzed thematically. This revealed five dominant themes that explain observed patterns and highlight residents’ perceptions and experiences.

- *The reservoir is declining* Residents believe that sedimentation and aging infrastructure have gradually reduced the reservoir’s surface water over time. One key informant described the reservoir as “like a dying battery; it holds less every year,” while another noted, “We lose more each year to silt,” indicating that recovery was unlikely as of 2013.

- *Wells Essential but Precarious.* Groundwater is considered vital, yet its long-term sustainability is uncertain. Participants stated, “*The wells saved us, but now we worry about them drying up, and “Some are already showing declining yields,”* indicating concerns about the actual capacity of the groundwater supply.
- *Springs Unreliable and seasonality* of spring yields are considered an inconsistent water source, heavily dependent on rainfall. One focus group participant explained, “*After good rain, they flow; during dry times, they disappear,”* while another added, “*springs are bonus water, not a base supply.”*
- *Increased supply, unmet needs* Although official data indicate an increase in water supply, many residents report no improvement in household water access. They commented, “*The numbers may show more water, but my tap still runs dry,”* and “*There are more people, but the same amount of water reaches us.”*
- *The water management system* is seen as responding to problems rather than preventing them. KIs remarked, “*We’re just getting by, not truly managing,”* and “*There isn’t a genuine plan; we’re merely reacting to emergencies,”* emphasizing the lack of implementing long-term strategic planning.

Themes emerged from inductive coding of six KIs and four FGDs, $n = 24$ participants). Theme 1 was mentioned by all 6 KIs and 12 FGD participants; Theme 2 by 5 KIs and 8 FGD participants; Theme 3 by 4 KIs and 16 FGD participants; Theme 4 by 0 KIs and 22 FGD participants; Theme 5 by 4 KIs and 18 FGD participants.

Thematic Network: The five themes form an interconnected causal network. The decline of the reservoir (Theme 1) drove diversification to groundwater (Theme 2), which created both relief and worry. This reliance on groundwater, combined with inadequate planning, reinforced reactive coping (Theme 5). Springs provide temporary seasonal relief (Theme 3) but remain unreliable. Most critically, the benefits of increased production fail to reach households (Theme 4), creating the perception-reality gap. Together, these interconnected themes characterize a system operating in perpetual “survival mode.”

Integrated Insight: The qualitative analysis reveals that Gondar’s water system operates in a perpetual reactive cycle. Physical decline of the reservoir (Theme 1) forced diversification to groundwater (Theme 2), but without adequate planning or monitoring (Theme 5), creating new vulnerabilities. Springs provide temporary relief but remain unreliable (Theme 3). Most critically, the benefits of increased production fail to reach households (Theme 4) due to distribution losses and population growth—a reality invisible in aggregate production statistics.

Discussion

The integrated statistical and qualitative analyses confirmed a fundamental reconfiguration of Gondar’s water supply architecture between 2011 and 2024. While the output from the Angereb Reservoir exhibited no significant long-term trend, groundwater abstraction and spring contributions increased significantly. Change point detection revealed a clear transition sequence, with reservoir decline beginning in July 2013, followed by groundwater expansion in July 2017, coinciding with Kola Diba wellfield commissioning and the integration of spring sources in June 2019, after rehabilitation programs.

These findings indicate that the observed growth in the total water supply reflects source substitution, rather than system expansion. The reservoir stagnated, whereas the wells and springs compensated for the stagnation. However, as qualitative evidence reveals, this substitution has not translated into improved household access, which is a critical finding for measuring and managing urban water security.

In practical terms, the increase in the overall water supply is due to source substitution, rather than system expansion. Specifically, wells and springs compensated for the reduced output of reservoirs. Therefore, because this substitution has not improved household access, management responses must focus on information systems, operational adjustments, infrastructure rehabilitation, loss reduction, and institutional governance rather than assuming that supply increases automatically lead to improved service delivery. Accordingly, this study proposes seven prioritized and actionable measures to strengthen urban water management.

1. *Establish a groundwater monitoring regime* This addresses the 12-month SPI response ($r = 0.432$), the lack of water level data, and the concerns about declining yields. Success will be measured by the installation of monitoring wells and the monthly reporting of water levels.
2. *Adopt a seasonal operations protocol* This accounts for the 11-month reservoir lag, two-month total water supply response time, and recurring winter scarcity. The indicator of success is the formal adoption of a seasonal operations plan and measurable reductions in dry season shortfalls.
3. *Plan and fund the rehabilitation of the Angereb Reservoir* This recovers storage lost to sedimentation and responds to the 2013 change-point and local accounts. Performance will be measured by the recovered storage capacity and improved dry season reliability.
4. *Formalize the spring sources* This responds to the three-month spring response ($r = 0.568$), the 2019 change-point, and high yield variability. The indicators include protected spring sites, consistent yield records, and integration with Supervisory Control and Data Acquisition (SCADA), where feasible.
5. *Reduce NRW* This narrows the perception versus reality gap and addresses estimated losses of 35 to 40 percent. Success is defined as a lower NRW percentage and improved continuity of supply at household taps.
6. *Develop a conjunctive use framework* This integrates distinct source response times and their substitution patterns. Progress will be demonstrated by the adoption of a conjunctive use plan and evidence of optimized source allocation during different seasonal phases of the project.
7. *Strengthening institutional coordination* This replaces reactive coping strategies with proactive management. Success indicators include a functioning coordination committee, jointly developed operational plans, and regular interagency reviews.

Prioritization rationale Recommended actions are ranked according to their urgency and practicality. Monitoring groundwater (Action 1) is the highest priority because current management decisions lack reliable data on

aquifer conditions. This absence of essential information is a major weakness, especially as cities increasingly depend on groundwater sources. Seasonal operations (Action 2) provide a quick and cost-efficient way to address climate-related mismatches revealed by lag analysis. Reservoir rehabilitation (Action 3) is crucial for ensuring long-term supply stability, but it requires substantial financial resources and technical preparation, making it a long-term project. Formalizing springs (Action 4) and reducing NRW (Action 5) focus on specific operational issues identified through qualitative analyses. Finally, planning for conjunctive use (Action 6), along with enhanced institutional coordination (Action 7), establishes the comprehensive governance needed to shift the system from reactive responses to proactive, integrated water management.

Evolution of water supply in Gondar city reflects the dynamic interactions between climatic forcing and institutional responses. Seasonal rainfall extremes interact with operational constraints to create complex adaptive pathways. The decline in surface water infrastructure triggered administrative responses, particularly the expansion of groundwater abstraction and the rehabilitation of spring sources. These measures helped stabilize the water supply in the short term but gradually shifted the vulnerability from surface water systems to subsurface resources.

Consequently, this feedback structure reveals strong path dependency. Delayed responses to reservoir sedimentation have postponed rehabilitation efforts and encouraged the reactive diversification of water sources. Although this strategy introduced resilience through redundancy, it lacked the integration that characterizes fully adaptive water management systems^{37,38}. Focus group participants repeatedly described the system as operating in “survival mode,” emphasizing that statistical increases in water production have not resulted in meaningful improvements in reliability or access to water.

The gap between perception and measured supply has significant implications for the assessment of urban water security. Conventional indicators that focus primarily on production volumes, the dominant metric in utility reporting, often conceal the distributional inequalities and demand-side challenges. In Gondar, aggregate statistics suggest that the supply has improved; however, household experiences reveal persistent, limited service. This finding supports the emerging critiques of production-centered water governance and strengthens calls for service-oriented indicators that better capture the reliability, accessibility, and equity of water provision^{39–41}.

Gondar's development trajectory resembles the patterns observed in other rapidly urbanizing regions. In Bengaluru, India, deep drilling into hard-rock aquifers temporarily sustained the urban water supply, but increased the long-term pressure on groundwater resources⁴². In Mexico City, extensive groundwater pumping has contributed to severe aquifer depletion and land subsidence⁴³, highlighting global concerns regarding unsustainable groundwater use⁴⁴. Gondar's highland hydrogeological conditions and rainfall-driven recharge provide a comparatively stronger potential for sustainable groundwater management, provided that systematic monitoring and adaptive management practices are implemented before critical thresholds are reached.

Conclusions

This study used a sequential explanatory mixed-methods approach to examine the long-term and seasonal dynamics of Gondar City's water supply from 2011 to 2024. Quantitative analysis showed that the output from the Angereb Reservoir remained largely stagnant, whereas groundwater abstraction and spring development expanded considerably, becoming the main drivers of overall supply growth. Despite this increase, seasonal scarcity persisted during winter and spring, reflecting a greater reliance on groundwater and uneven distribution within the supply system. Change point detection identified several regime shifts, including declining reservoir storage in 2013, groundwater expansion in 2017, and spring water integration in 2019. Qualitative evidence supports these findings, linking the identified shifts to documented infrastructure interventions and climate-related events. Lagged correlation analysis further revealed distinct climate response times for different sources: an inverse response of 11 to 12 months for the reservoir, three months for springs, 12 months for groundwater, and two months for total supply.

These findings suggest an asymmetric adaptation pathway, whereby diversification toward subsurface sources offsets the stagnation of the surface water supply. Although this strategy maintained total production, it introduced vulnerabilities by increasing dependence on largely unregulated groundwater abstractions. Notably, qualitative evidence indicates that statistical increases in the total supply have not translated into improved household access, a discrepancy that can be attributed to rapid population growth, losses within the distribution network, and uneven coverage of water infrastructure across the city.

Achieving sustainable urban water security in Gondar requires a transition from reactive coping to proactive and integrated management strategies. Key priorities include establishing groundwater monitoring systems, introducing seasonal operational protocols based on source-specific response times, rehabilitating the Angereb Reservoir, formalizing and protecting spring sources, reducing non-revenue water, developing conjunctive use strategies that balance surface and groundwater resources, and strengthening coordination among responsible institutions.

Future research can build on this study by addressing these limitations. First, well-specific monitoring is required to assess groundwater drawdown patterns and determine sustainable extraction levels. Second, household-level surveys should be conducted to quantify inequalities in water access and distribution. Third, optimization modeling can help determine efficient seasonal allocation strategies under climate variability. Fourth, comparative studies across other Ethiopian highland cities would help evaluate the broader applicability of the patterns observed in this study. Finally, integrating downscaled climate projections would allow the assessment of system vulnerability under projected conditions for the period 2030–2050.

Overall, this study demonstrates the value of mixed-methods socio-hydrological analysis for understanding the physical and institutional drivers of resilience in rapidly urbanizing and resource-constrained environments in the Global South. By combining statistical evidence with stakeholder perspectives, this study provides a framework for diagnosing urban water supply transitions and informing context-appropriate policy responses.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Monthly water supply time series and qualitative interview transcripts (anonymized) are available for research purposes, subject to institutional data sharing agreements.

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Author contributions

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Competing interests

The authors declare that they have no competing interests.

Generative AI and AI-assisted technologies

During the preparation of this work, the authors used AI-based programs, including ChatGPT, Grammarly, and QuillBot, for language editing and proofreading to enhance readability and correct grammatical errors. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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